KISSsoft 2026 native report - readable text rendering
Source: R7-bearing-Fa300.pprpt
No recalculation. Graphic payloads omitted; retain native PPRPT for full layout.

Rolling bearing calculation according to
 ISO 76 & ISO 281
R7-bearing-Fa300
Changed on | 31.10.2026, 16:52:53 |
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Description | KISSsoft example |
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 | 2o2dy (5416) |
 | KISSsoft Release 2026 |
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1 Messages
| Calculation is consistent. |
2 Overview
Calculation method: ISO 281:2007 and manufacturer's notes
- With constant a23-factor (1.0)
General data:
Speed (1/min) 900.000
Axial force (N) 300.000
Required service life (h) 5000.000
2.1 Rolling bearing 1
Designation SKF 6206-RZ
Type Deep groove ball bearing (single row)
SKF Explorer
Bearing clearance: C0
Radial and axial load
Radial force (N) [Fr] 664.917
Axial force (N) [Fa] 300.000
Inner diameter (mm) [d] 30.000
External diameter (mm) [D] 62.000
Width (mm) [B] 16.000
Basic dynamic load rating (kN) [C] 20.300
Basic static load rating (kN) [C0] 11.200
Fatigue load limit (kN) [Cu] 0.475
Dynamic equivalent load (N) [P] 949.853
2.1.1 Results of bearing calculation according to ISO 281:2007
Speed limit (oil) (1/min) [nmax] 12000
Static equivalent load (N) [P0] 664.917
Basic rating life (h) [L10h] 180769.261
Bearing rating life (h) [L10h] 180769.261
Static safety factor [S0] 16.844
2.1.2 Friction
Rolling moment of friction (Nmm) [Mrr] 35.076
Sliding moment of friction (Nmm) [Msl] 8.195
Moment of friction, seals (Nmm) [Mseal] 0.000
Mseal according to SKF Main Catalog 17000/1 EN: 2018
Moment of friction flow losses (Nmm) [Mdrag] 0.000
Total moment of friction (Nmm) [M] 43.271
2.2 Rolling bearing 2
Designation SKF 6206-RZ
Type Deep groove ball bearing (single row)
SKF Explorer
Bearing clearance: C0
Only radial load
Radial force (N) [Fr] 664.920
Axial force (N) [Fa] 0.000
Inner diameter (mm) [d] 30.000
External diameter (mm) [D] 62.000
Width (mm) [B] 16.000
Basic dynamic load rating (kN) [C] 20.300
Basic static load rating (kN) [C0] 11.200
Fatigue load limit (kN) [Cu] 0.475
Dynamic equivalent load (N) [P] 664.920
2.2.1 Results of bearing calculation according to ISO 281:2007

Speed limit (oil) (1/min) [nmax] 12000
Static equivalent load (N) [P0] 664.920
Basic rating life (h) [L10h] 526969.637
Bearing rating life (h) [L10h] 526969.637
Static safety factor [S0] 16.844
2.2.2 Friction
Rolling moment of friction (Nmm) [Mrr] 14.279
Sliding moment of friction (Nmm) [Msl] 3.094
Moment of friction, seals (Nmm) [Mseal] 0.000
Mseal according to SKF Main Catalog 17000/1 EN: 2018
Moment of friction flow losses (Nmm) [Mdrag] 0.000
Total moment of friction (Nmm) [M] 17.373
3 Remarks
The moment of friction M is calculated according to the details in the SKF Catalog 2018.
This is always calculated with a coefficient for additives in the lubricant mybl=0.15.
End of report (lines: 122)